CE432
TERM PAPER

USING OPEN DATABASES TO DEVELOP GIS APPLICATION (E.G. OPENSTREET MAP.)

IITK NAVIGATOR PRO: A GRAPH-BASED MULTI-MODAL CAMPUS ROUTING SYSTEM

USING OPEN DATABASES TO DEVELOP GIS APPLICATION (E.G. OPENSTREET MAP.)

GROUP-9

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INTRODUCTION

- Large institutional campuses have complex spatial layouts
- Navigation is primarily pedestrian and cycle-based
- Internal pathways are often not well represented in standard maps

PROBLEM CONTEXT

- Conventional systems (e.g., Google Maps):
  - Focus on vehicular navigation
  - Do not capture internal pathways
- Results in:
  - Incomplete routes
  - Last-mile navigation problem

OBJECTIVES

- Extract spatial data from open-source platforms
- Construct graph-based campus network
- Enable multi-modal routing (walk, cycle, drive)
- Generate turn-by-turn navigation
- Develop an interactive GIS-based application

BACKGROUND WORK

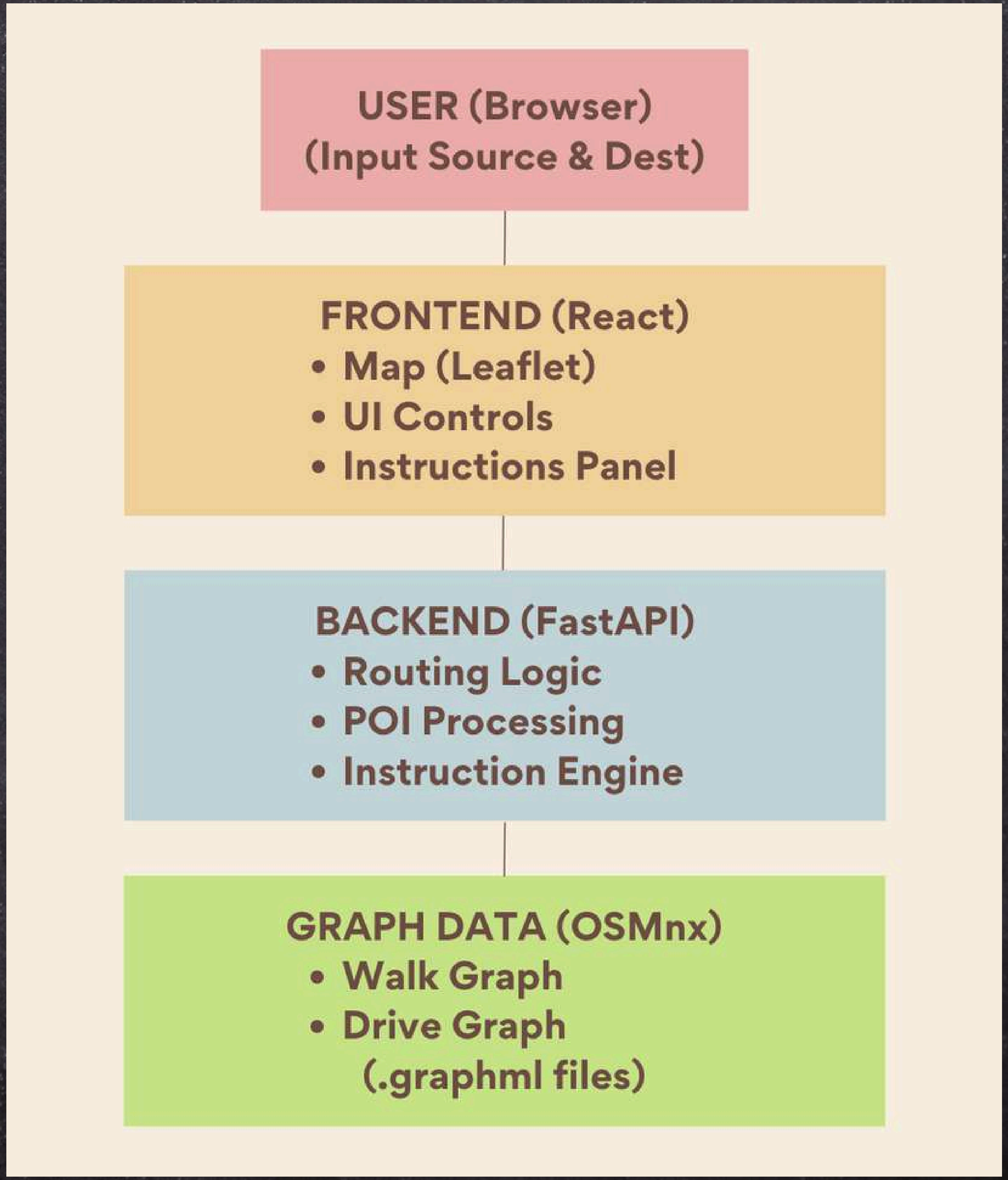
- GIS routing systems are based on graph theory
- Spatial networks modeled as:
  - Nodes → intersections
  - Edges → pathways
- OpenStreetMap provides detailed, tagged geographic data

TOOLS AND TECH STACK

- **OSMnx** → spatial data extraction
- **NetworkX** → graph construction & analysis
- **Dijkstra's Algorithm** → shortest path
- **React + Leaflet** → frontend visualization
- **FastAPI** → backend services

METHODOLOGY OVERVIEW

- Data extraction from OSM
- Graph generation
- Routing computation
- Instruction generation
- Web-based visualization



DATA COLLECTION & GRAPH GENERATION

- Defined a 1500 m boundary around IIT Kanpur
- Extracted spatial data using OSMnx
- Generated two separate graphs:
    - Walk/Cycle graph (footpaths + roads)
    - Drive graph (roads only)
- Stored as .graphml format

POI EXTRACTION

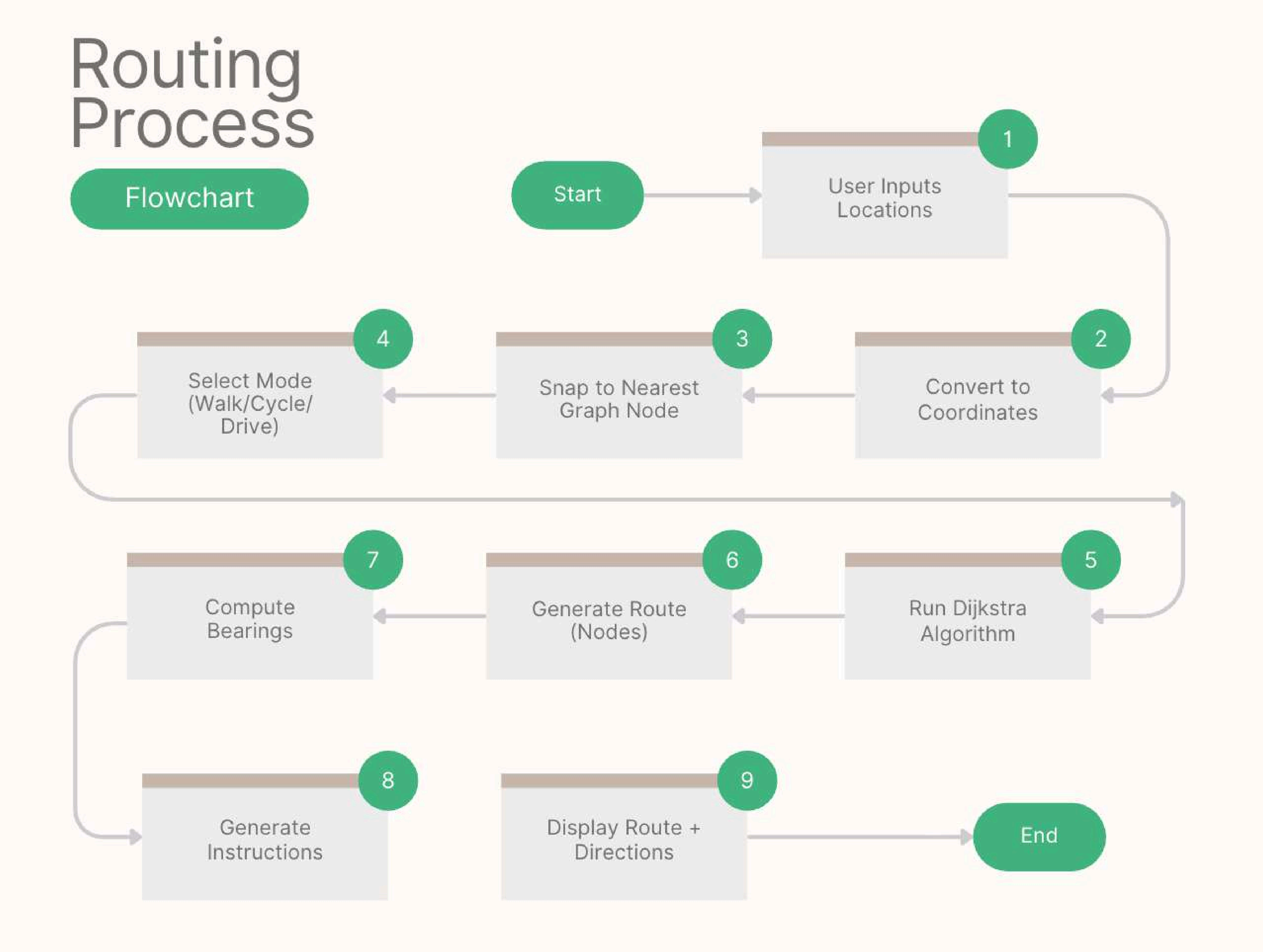
- Identified buildings and amenities from OSM tags
- Computed centroid for each location
- Categorized into:
    - Food & Cafés
    - ATMs
    - Healthcare
- Stored for efficient querying

ROUTING ALGORITHM

- User input converted to geographic coordinates
- Coordinates mapped to nearest graph nodes
- Shortest path computed using Dijkstra's algorithm
- Travel time estimated using predefined speeds

ROUTING PROCESS

- Input locations
- Node mapping
- Path computation
- Route visualization
- Instruction generation



TURN-BY-TURN DIRECTIONS

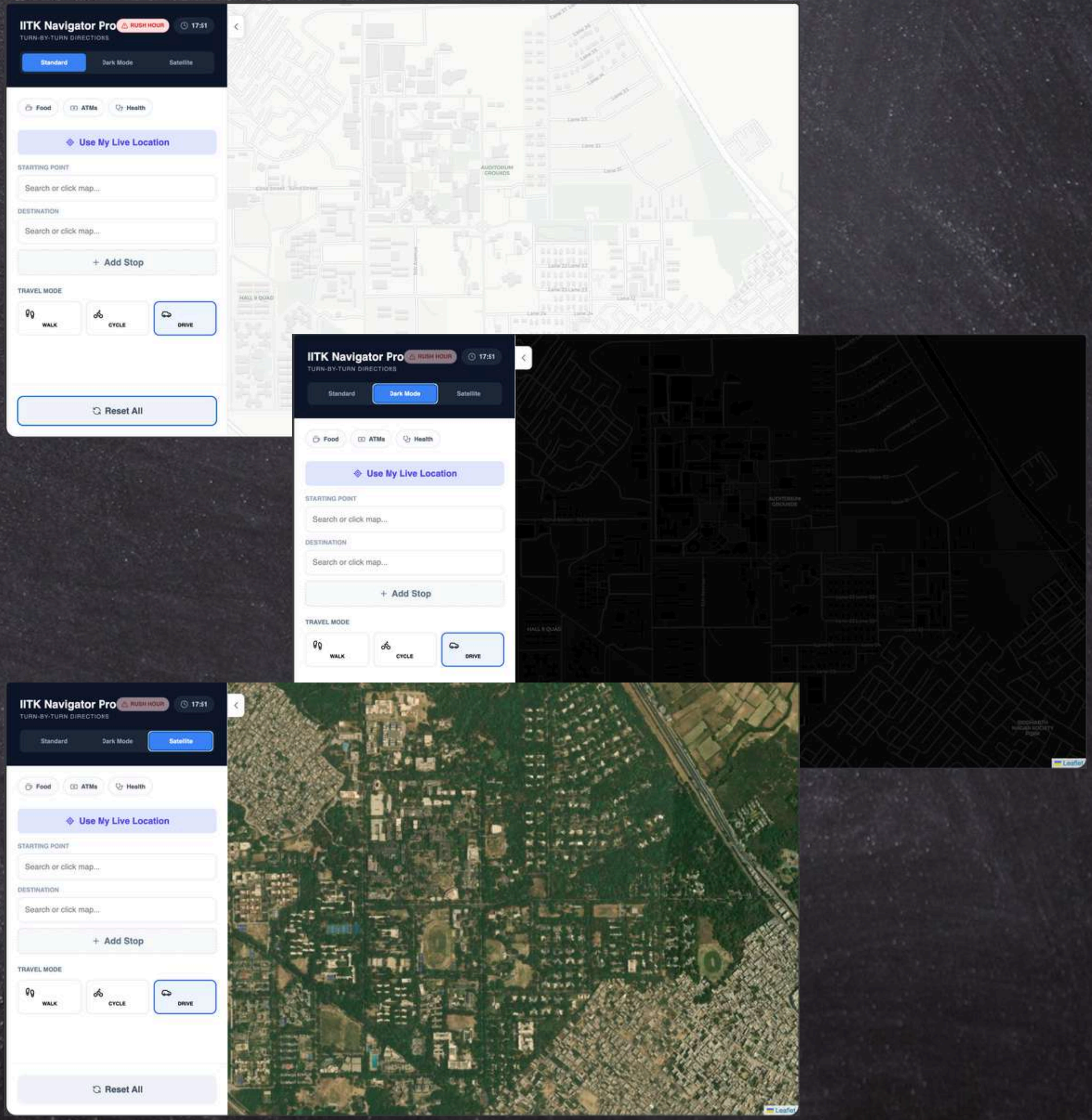
- Bearings calculated between consecutive nodes
- Angular differences used to determine direction
- Generates instructions such as:
    - Turn left
    - Turn right
    - Continue straight

SYSTEM IMPLEMENTATION

- Frontend developed using React and Leaflet
- Backend implemented using FastAPI
- Communication via REST APIs
- Deployed on web platform

USER INTERFACE FEATURES

- Interactive map-based interface
- Multiple visualization modes:
  - Standard
  - Dark Mode
  - Satellite
- Real-time user interaction



RESULTS

MODE-SPECIFIC ROUTING

Feature	Walk/Cycle	Drive
Paths Used	Footpaths + Roads	Only Roads
Route Length	Shorter	Longer
Accessibility	High	Limited
Last-Mile Accuracy	Exact building reach	Road-side drop-off
Use Case	Students walking/cycling	Vehicles

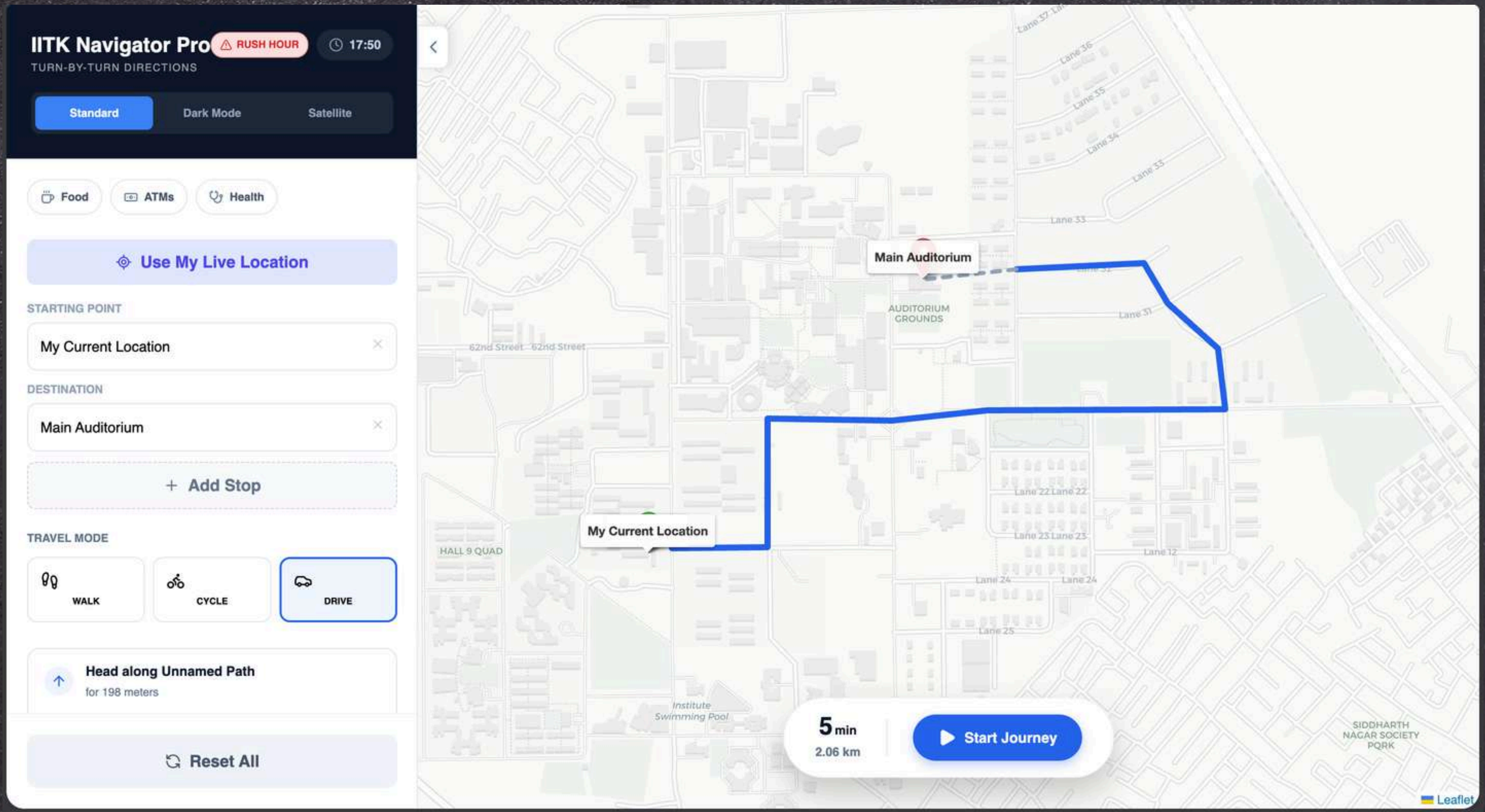
- Walking routes use internal campus pathways
- Driving routes restricted to road network
- Improved routing accuracy and usability

LAST-MILE NAVIGATION

- Combines road network with internal access paths
- Provides route up to exact building location
- Solves last-mile navigation issue

RESULTS

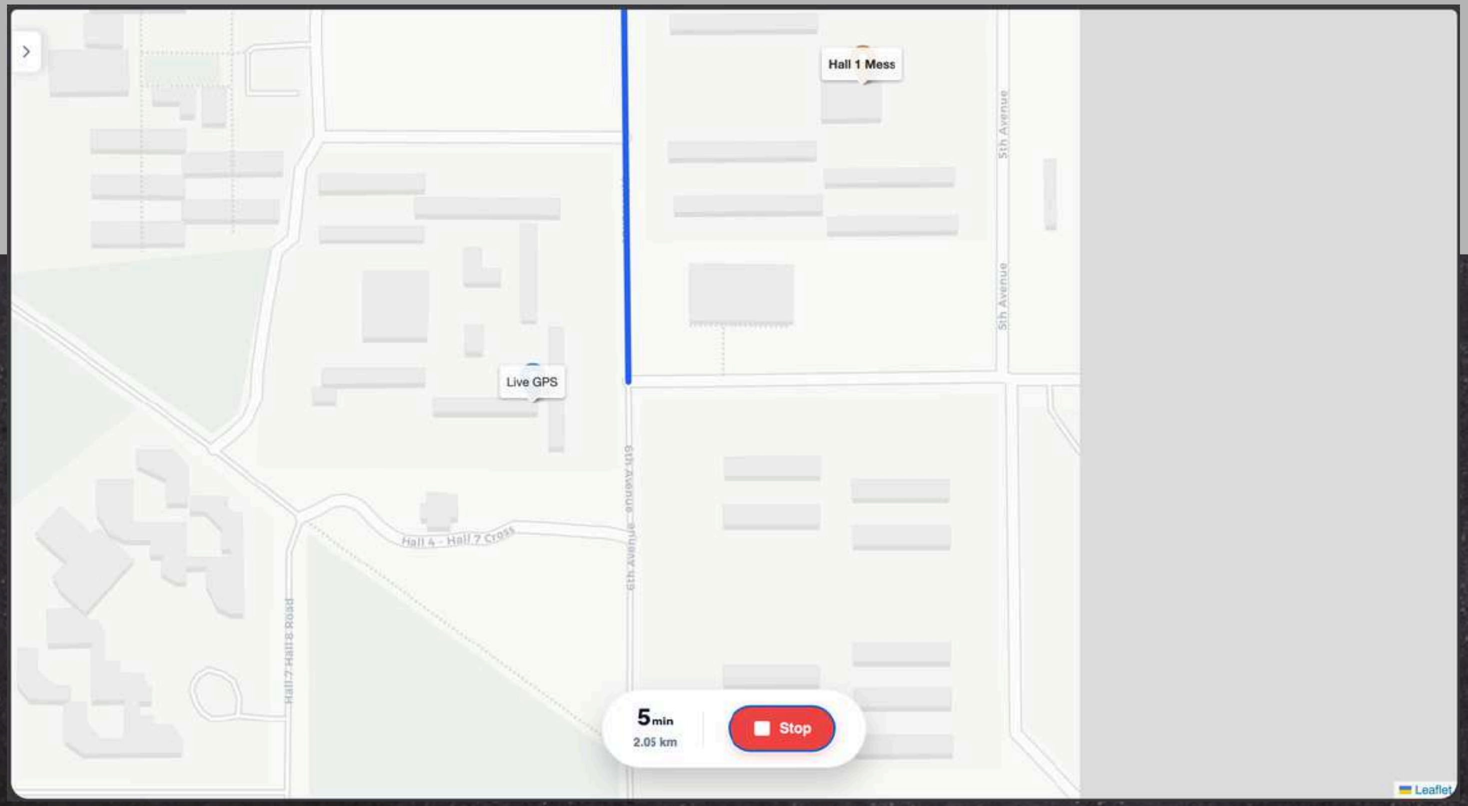
NAVIGATION OUTPUT



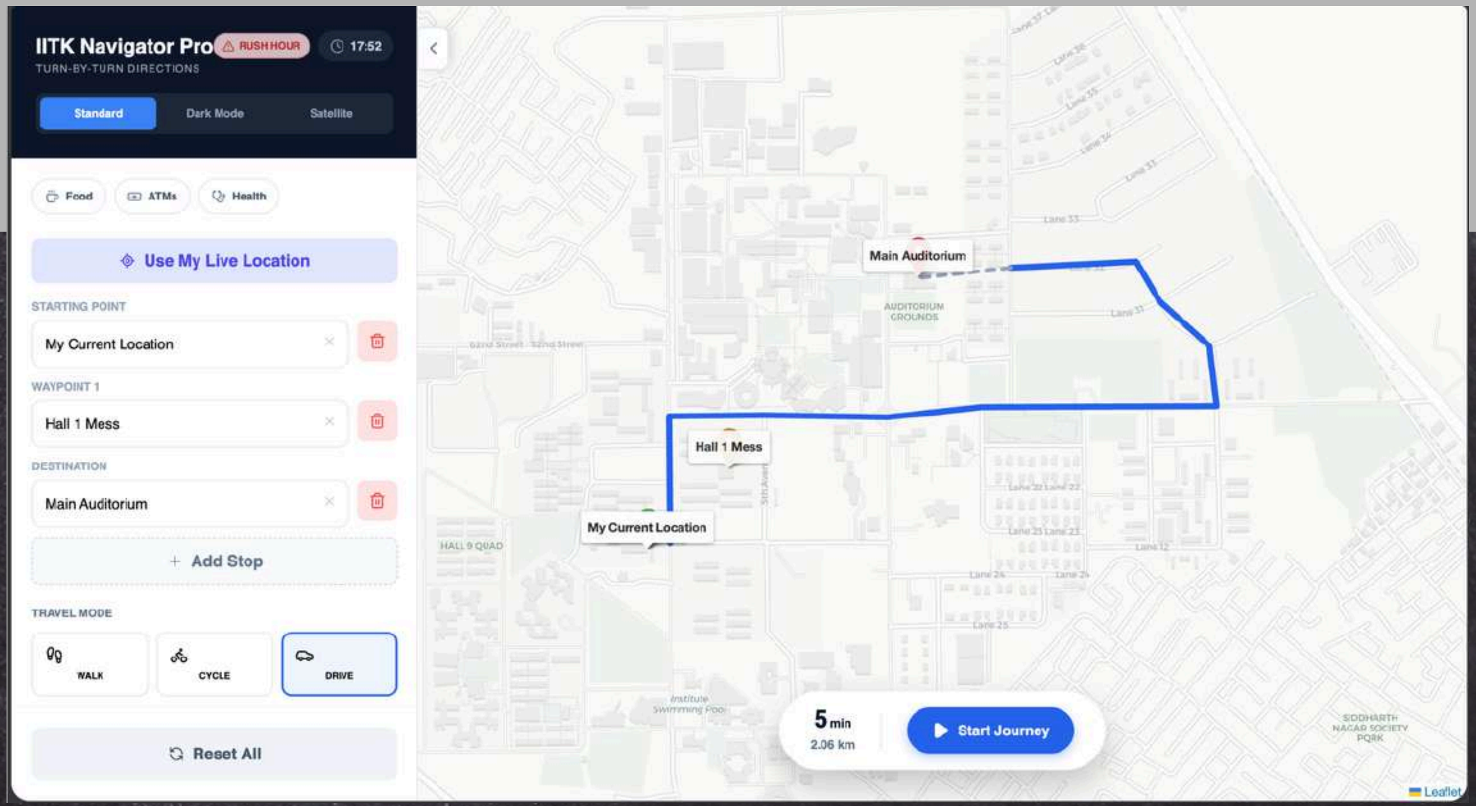
- Route displayed on map
- Distance and estimated time shown
- Step-by-step navigation instructions generated

ADDITIONAL FEATURES

LIVE LOCATION TRACKING

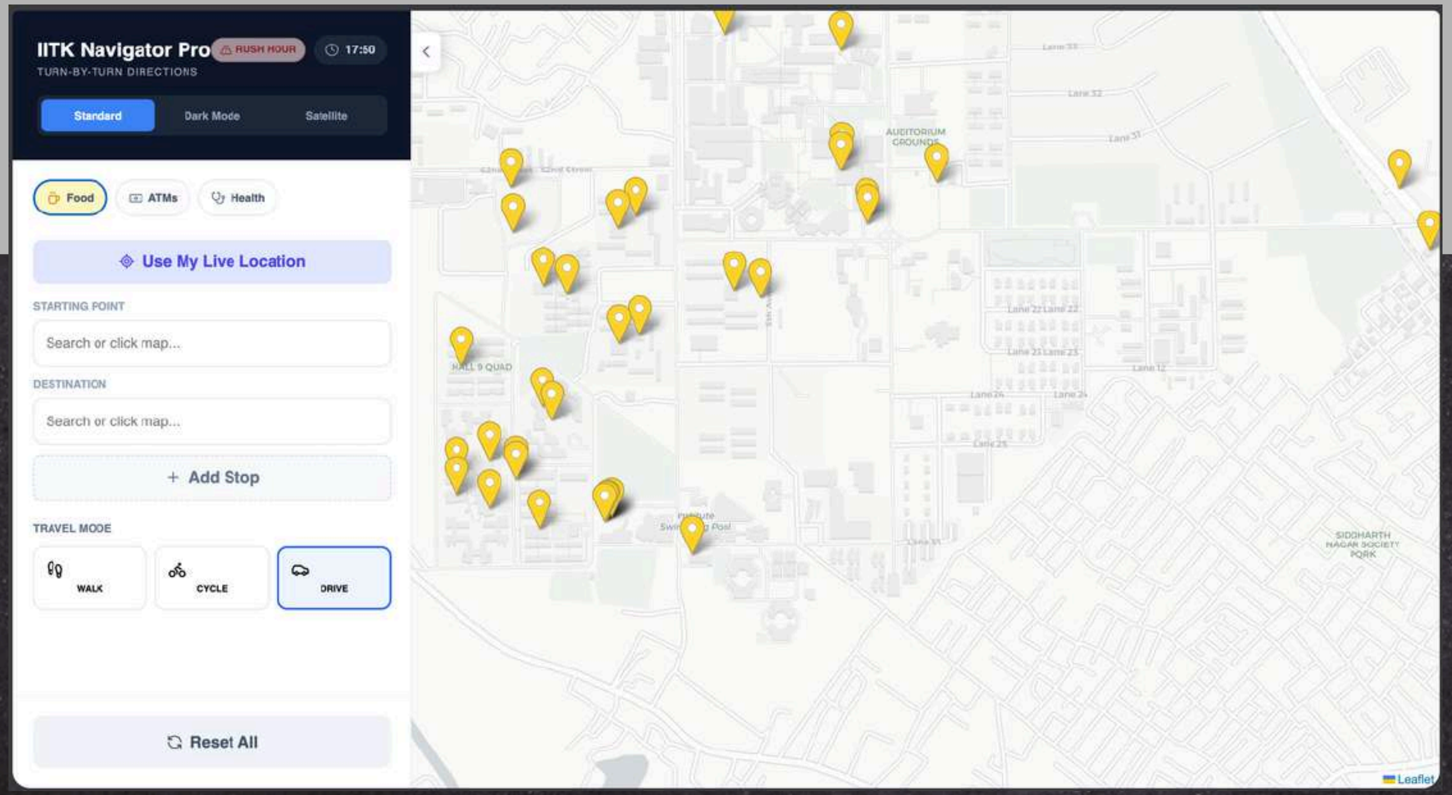


MULTI-STOP ROUTING CAPABILITY

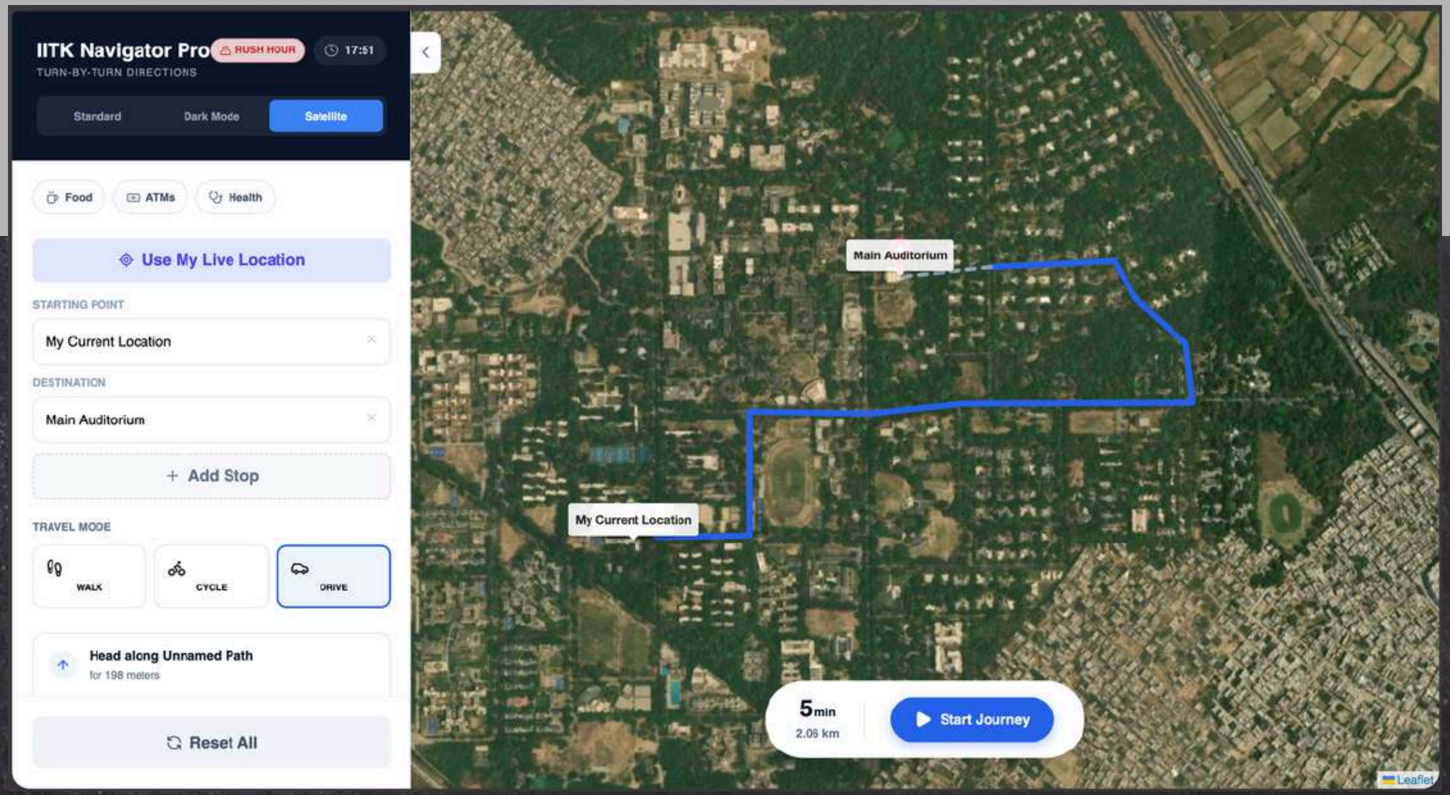


ADDITIONAL FEATURES

AMENITY VISUALIZATION ON MAP



RUSH HOUR INDICATION



TIME-AWARE ROUTING

- Identifies predefined rush hour periods
- Applies increased weight to edges (~2.5×)
- Improves realism of travel time estimation

KEY INSIGHTS

- Custom graph-based models improve accuracy
- Separation of travel modes enhances usability
- Integration of temporal factors improves routing realism

CHALLENGES

- Incomplete or missing OSM data
- Ambiguity in path naming
- Handling large graph structures
- Memory and performance constraints

LIMITATIONS

- Dependence on OSM data quality
- No real-time traffic integration
- Heuristic-based congestion modeling
- Limited accessibility features

CONCLUSION

- Developed a GIS-based campus navigation system
- Successfully addressed last-mile navigation
- Demonstrated practical use of open GIS data

FUTURE SCOPE

- Accessibility-aware routing
- Integration of real-time data
- Mobile application development
- Isochrone-based analysis

THANK YOU!